PROBABILITY vs STATISTICS — CONCEPTUAL MCQs

1. You know that a fair die has six equally likely outcomes. You want to calculate the probability of getting a 6. This is an example of: A. Statistics B. Probability C. Regression D. Sampling

Answer : B Probability

2. You roll a die 1,000 times and record the results. You notice that the number 6 appeared much more frequently than expected. You want to determine whether the die may be biased. This is primarily: A. Probability B. Statistics C. Linear Algebra D. Calculus

Answer : B Statistics

3. Which statement best describes Probability? A. Using observed data to estimate unknown characteristics B. Using a known model or assumptions to reason about possible outcomes C. Organizing data into tables D. Calculating averages from collected data

Answer : B Using a known model or assumptions to reason about possible outcomes

4. Which statement best describes Statistics? A. Predicting possible outcomes from a known probability model B. Using observed data to understand or estimate an underlying process C. Generating random numbers D. Calculating only probabilities between 0 and 1

Answer : B Using observed data to understand or estimate an underlying process

5. A company knows that historically 2% of its products are defective. It wants to estimate how many defective products might appear in the next 10,000 products. This is primarily: A. Statistics B. Probability C. Data cleaning D. Visualization

Answer : B Probability

6. A company collects data from 10,000 products and finds that 250 are defective. It wants to estimate the defect rate of the production process. This is primarily: A. Probability B. Statistics C. Linear Algebra D. Optimization

Answer : B Statistics

7. Which direction best represents Probability? A. Data → Model B. Model → Possible outcomes C. Data → Visualization D. Sample → Database

Answer : B Model → Possible outcomes

8. Which direction best represents Statistics? A. Model → Possible outcomes B. Possible outcomes → Model C. Observed data → Understanding/inference about the underlying process D. Probability → Randomness

Answer : C Observed data → Understanding/inference about the underlying process

9. A medical researcher knows that a disease occurs in approximately 1% of the population. She wants to calculate the probability that a randomly selected person has the disease. This is primarily: A.

Statistics B. Probability C. Data preprocessing D. Regression

Answer : B Probability

10. A medical researcher observes that 15 out of 1,000 randomly selected people have a particular disease. She wants to estimate the disease prevalence in the larger population. This is primarily: A. Probability B. Statistics C. Matrix algebra D. Optimization

Answer : B Statistics

11. Consider the following situation: "A coin is believed to be fair. What is the probability of getting 7 Heads when the coin is tossed 10 times?" This is: A. Statistics B. Probability C. Descriptive statistics D. Sampling

Answer : B Probability

12. Now consider: "A coin is tossed 10 times and produces 9 Heads. Based on this observation, determine whether the coin is likely to be fair." This is: A. Probability B. Statistics C. Matrix multiplication D. Data visualization

Answer : B Statistics

13. Which question is a Probability question? A. What is the average height of students in this class? B. What is the probability that a randomly selected student is taller than 180 cm? C. What is the median height of the students? D. What is the standard deviation of student heights?

Answer : B What is the probability that a randomly selected student is taller than 180 cm?

14. Which question is a Statistics question? A. What is the probability of getting Heads on a fair coin? B. What is the probability of rolling a 6? C. What is the average income observed in this dataset? D. What is the probability of rain tomorrow?

Answer : C What is the average income observed in this dataset?

15. A weather model says that there is a 70% probability of rain tomorrow. This statement is an example of: A. Statistics B. Probability C. Descriptive statistics D. Sampling

Answer : B Probability

16. You collect the rainfall measurements from the last 20 years and calculate the average annual rainfall. This activity is primarily: A. Probability B. Statistics C. Probability distribution D. Random sampling

Answer : B Statistics

Just to make it clearer – Probability and statistics

17. "What is likely to happen?" A. Probability B. Statistics

Answer : A Probability

18. "What happened in the data?" A. Probability B. Statistics

Answer : B Statistics

19. "What can we learn about the population from our sample?" A. Probability B. Statistics

Answer : B Statistics

20. "If the assumptions are known, what outcomes should we expect?" A. Probability B. Statistics

Answer : A Probability